

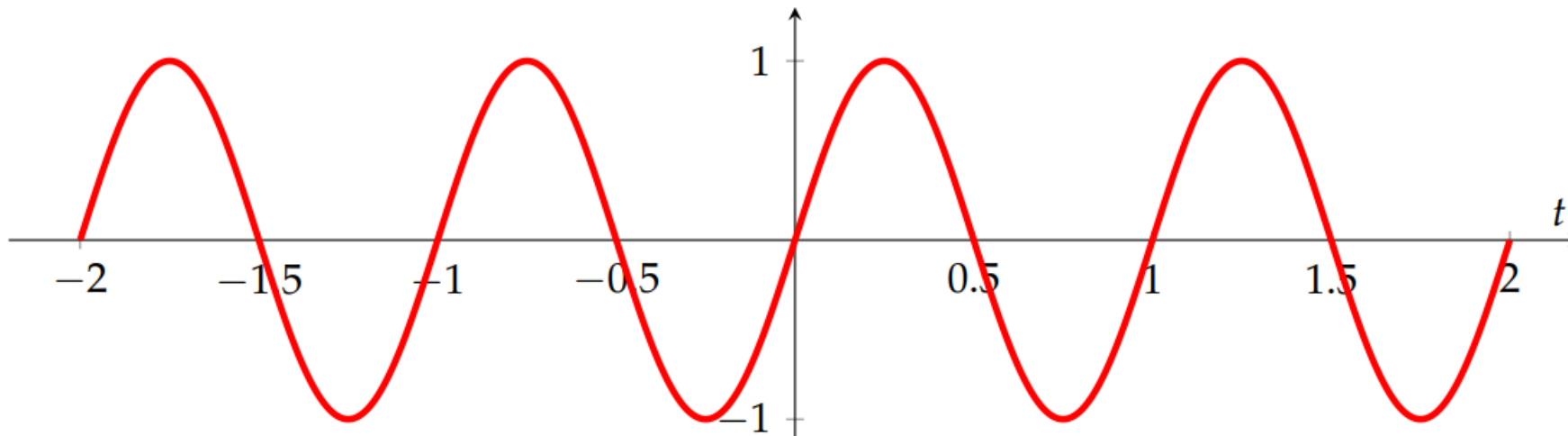


FREKVENCIJSKE KARAKTERISTIKE

SISTEMI AUTOMATSKOG UPRAVLJANJA
RAČUNARSKE VEŽBE

U SPEKTRALNOJ ANALIZI SVE SIGNALA RAZLAŽEMO NA ELEMENTARNE,
PROSTOPERIODIČNE OBLIKE.

**OSNOVNI GRADIVNI ELEMENT SIGNALA SU
PROSTOPERIODIČNI TALASI.**

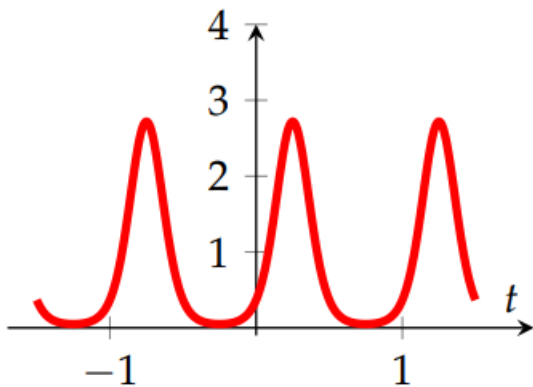


PERIODIČNI SIGNALI

Signal f je periodičan sa **periodom** $T > 0$ ukoliko za svaki vremenski trenutak t važi

$$f(t) = f(t+T)$$

$$f(t) = f(t \pm T) = f(t \pm 2T) = f(t \pm 3T) = \dots$$



PROSTOPERIODIČNI SIGNALI

Prostoperiodičan signal osnovnog perioda $T > 0$ je svaki signal f koji se može zapisati u jednom od sledećih ekvivalentnih oblika:

$$f(t) = a \cos\left(\frac{2\pi}{T}t\right) + b \sin\left(\frac{2\pi}{T}t\right)$$

$$f(t) = A \cos\left(\frac{2\pi}{T}t + \varphi\right)$$

$$f(t) = A \sin\left(\frac{2\pi}{T}t + \Phi\right)$$

a, b – doprinosi

A – amplituda

φ, Φ – fazni stavovi

PROSTOPERIODIČNI SIGNALI

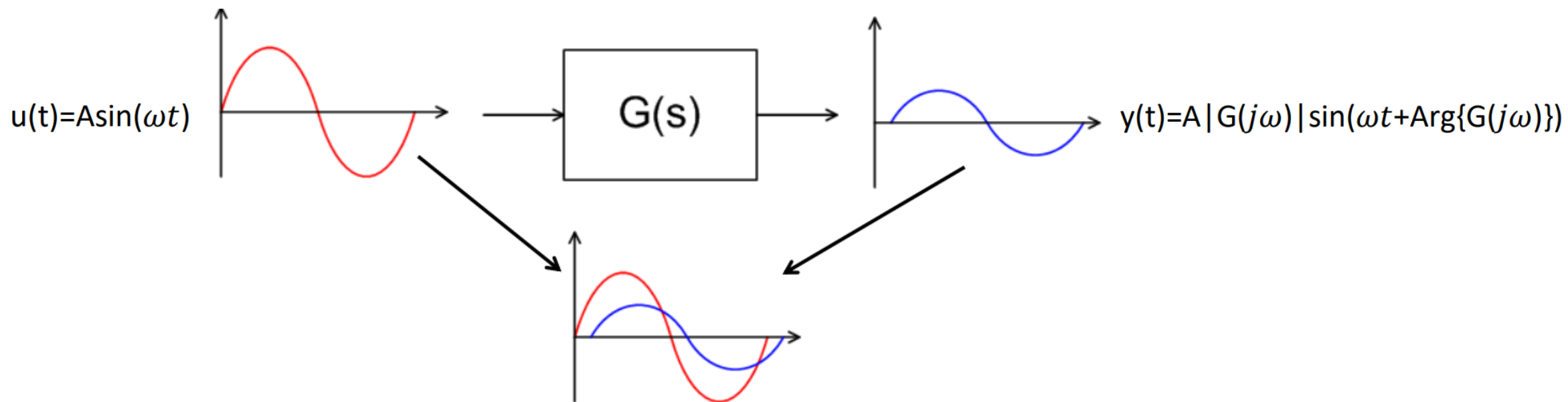
Broj punih perioda u jediničnom intervalu vremena naziva se **učestanošću** (tj. **frekvencijom**) signala. Ona je recipročna brojnoj vrednosti osnovnog perioda:

$$\nu = \frac{1}{T}$$

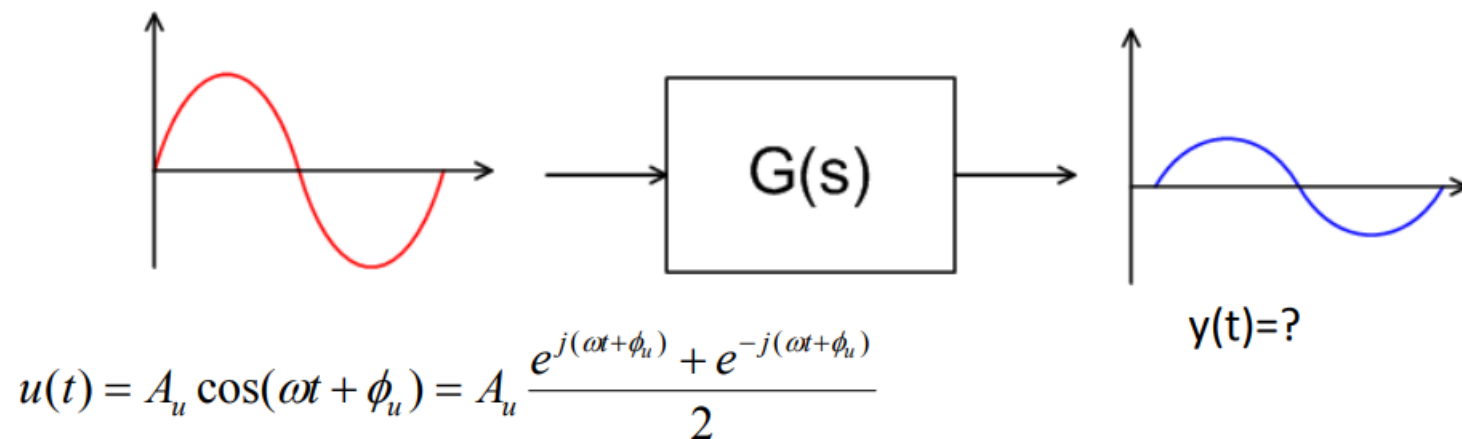
Kružna učestanost signala jednaka je ukupnoj brzini promene argumenta prostoperiodičnog signala, a proporcionalna je učestanosti:

$$\omega = 2\pi\nu = \frac{2\pi}{T}$$

KADA SE NA ULAZ LINEARNOG SISTEMA DOVEDE PROSTOPERIODIČNI SIGNAL, NA IZLAZU SE DOBIJA TAKOĐE PROSTOPERIODIČNI SIGNAL, ISTE UČESTANOSTI SAMO SA PROMENJENOM AMPLITUDOM I FAZOM.



FREKVENCIJSKE KARAKTERISTIKE



Pretpostavka: sistem je stabilan

$$Y(s) = G(s)U(s) \quad y(t) = \int_0^t g(\tau)u(t-\tau)d\tau = \int_0^\infty g(\tau)u(t-\tau)d\tau - \int_t^\infty g(\tau)u(t-\tau)d\tau$$

$$\int_0^\infty g(\tau)u(t-\tau)d\tau \quad - \text{ustaljeno stanje}$$

$$\int_t^\infty g(\tau)u(t-\tau)d\tau \quad - \text{prelazni režim}$$

FREKVENCIJSKE KARAKTERISTIKE

Odziv u ustaljenom stanju:

$$y(t) = \int_0^{\infty} g(\tau) u(t-\tau) d\tau = \int_{-\infty}^{\infty} g(\tau) u(t-\tau) d\tau = \int_{-\infty}^{\infty} g(\tau) A_u \frac{e^{j(\omega(t-\tau)+\phi_u)} + e^{-j(\omega(t-\tau)+\phi_u)}}{2} d\tau$$

$$y(t) = \frac{A_u}{2} \int_{-\infty}^{\infty} g(\tau) e^{j(\omega(t-\tau)+\phi_u)} d\tau + \frac{A_u}{2} \int_{-\infty}^{\infty} g(\tau) e^{-j(\omega(t-\tau)+\phi_u)} d\tau$$

$$y(t) = \frac{A_u}{2} e^{j(\omega t + \phi_u)} \int_{-\infty}^{\infty} g(\tau) e^{-j\omega\tau} d\tau + \frac{A_u}{2} e^{-j(\omega t + \phi_u)} \int_{-\infty}^{\infty} g(\tau) e^{j\omega\tau} d\tau$$

$$G(j\omega) = \int_{-\infty}^{\infty} g(\tau) e^{-j\omega\tau} d\tau \quad \text{F-ja prenosa u frekvencijskom domenu}$$

$$G(-j\omega) = \int_{-\infty}^{\infty} g(\tau) e^{j\omega\tau} d\tau :$$

FREKVENCIJSKE KARAKTERISTIKE

$$y(t) = \frac{A_u}{2} e^{j(\omega t + \phi_u)} G(j\omega) + \frac{A_u}{2} e^{-j(\omega t + \phi_u)} G(-j\omega)$$

$$y(t) = \frac{A_u}{2} e^{j(\omega t + \phi_u)} |G(j\omega)| e^{j \text{Arg}\{G(j\omega)\}} + \frac{A_u}{2} e^{-j(\omega t + \phi_u)} |G(-j\omega)| e^{j \text{Arg}\{G(-j\omega)\}}$$

$$y(t) = \frac{A_u}{2} e^{j(\omega t + \phi_u)} |G(j\omega)| e^{j \text{Arg}\{G(j\omega)\}} + \frac{A_u}{2} e^{-j(\omega t + \phi_u)} |G(j\omega)| e^{-j \text{Arg}\{G(j\omega)\}}$$

$$y(t) = |G(j\omega)| A_u \left(\frac{1}{2} e^{j(\omega t + \phi_u + \text{Arg}\{G(j\omega)\})} + \frac{1}{2} e^{-j(\omega t + \phi_u + \text{Arg}\{G(j\omega)\})} \right)$$

$$y(t) = |G(j\omega)| A_u \cos(\omega t + \phi_u + \text{Arg}\{G(j\omega)\}) = A_i \cos(\omega t + \phi_i)$$

$$|G(j\omega)| = \frac{A_i}{A_u} \quad - \text{pojačanje}$$

$$\text{Arg}\{G(j\omega)\} = \phi_i - \phi_u \quad - \text{fazno pomeranje}$$

ZADATAK 1.

Dat je sistem opisan funkcijom prenosa $G(s) = \frac{1}{s+1}$. Na ulaz sistema se dovodi signal $u(t) = 2\sin(3t)$. Proceniti vrednost signala na izlazu u ustaljenom stanju.

$$G(j\omega) = \frac{1}{j\omega+1}$$

$$y(t) = 2|G(3j)|\sin(3t + \text{Arg}\{G(3j)\})$$

$$|G(j\omega)| = \frac{1}{\sqrt{\omega^2 + 1^2}} = \frac{1}{\sqrt{\omega^2 + 1}}$$

$$|G(3j)| = \frac{1}{\sqrt{3^2 + 1^2}} = \frac{1}{\sqrt{10}} = 0.3162$$

$$\text{Arg}\{G(j\omega)\} = 0 - \arctg \frac{\omega}{1} = -\arctg(\omega)$$

$$\text{Arg}\{G(3j)\} = -\arctg(3) = -1.249$$

$$y(t) = 2 \cdot 0.3162 \sin(3t - 1.249) = 0.6324 \sin(3t - 1.249)$$

ZADATAK 2.

Sistem je opisan funkcijom prenosa $G(s) = \frac{s+1}{s+2}$. Proceniti signal na izlazu sistema u ustaljenom stanju ukoliko je pobudni signal $u(t) = 2\cos(t - \frac{\pi}{2})$.

$$G(s) = \frac{j\omega + 1}{j\omega + 2}$$

$$y(t) = 2|G(1j)|\cos(t - \frac{\pi}{2} + \text{Arg}\{G(1j)\})$$

$$|G(j\omega)| = \frac{\sqrt{\omega^2 + 1}}{\sqrt{\omega^2 + 4}}$$

$$|G(1j)| = \frac{\sqrt{1^2 + 1}}{\sqrt{1^2 + 4}} = \frac{\sqrt{2}}{\sqrt{5}} = 0.6324$$

$$\text{Arg}\{G(j\omega)\} = \arctg(\omega) - \arctg \frac{\omega}{2}$$

$$\text{Arg}\{G(1j)\} = \arctg(1) - \arctg \frac{1}{2} = 0.3217$$

$$y(t) = 2 \cdot 0.6324 \cos\left(t - \frac{\pi}{2} + 0.3217\right) = 1.2648 \cos(t - 1.2491)$$

ZADATAK 3.

Dat je sistem opisan funkcijom prenosa $G(s) = \frac{1}{s+1}$. Na ulaz sistema se dovodi signal $u(t) = \sin(2t + 3) + 10 \cos(4t - 2)$. Proceniti vrednost signala na izlazu u ustaljenom stanju.

$$G(j\omega) = \frac{1}{j\omega+1}$$

$$y(t) = |G(2j)|\sin(2t + 3 + \text{Arg}\{G(2j)\}) + 10|G(4j)|\cos(4t - 2 + \text{Arg}\{G(4j)\})$$

$$|G(j\omega)| = \frac{1}{\sqrt{\omega^2 + 1^2}} = \frac{1}{\sqrt{\omega^2 + 1}}$$

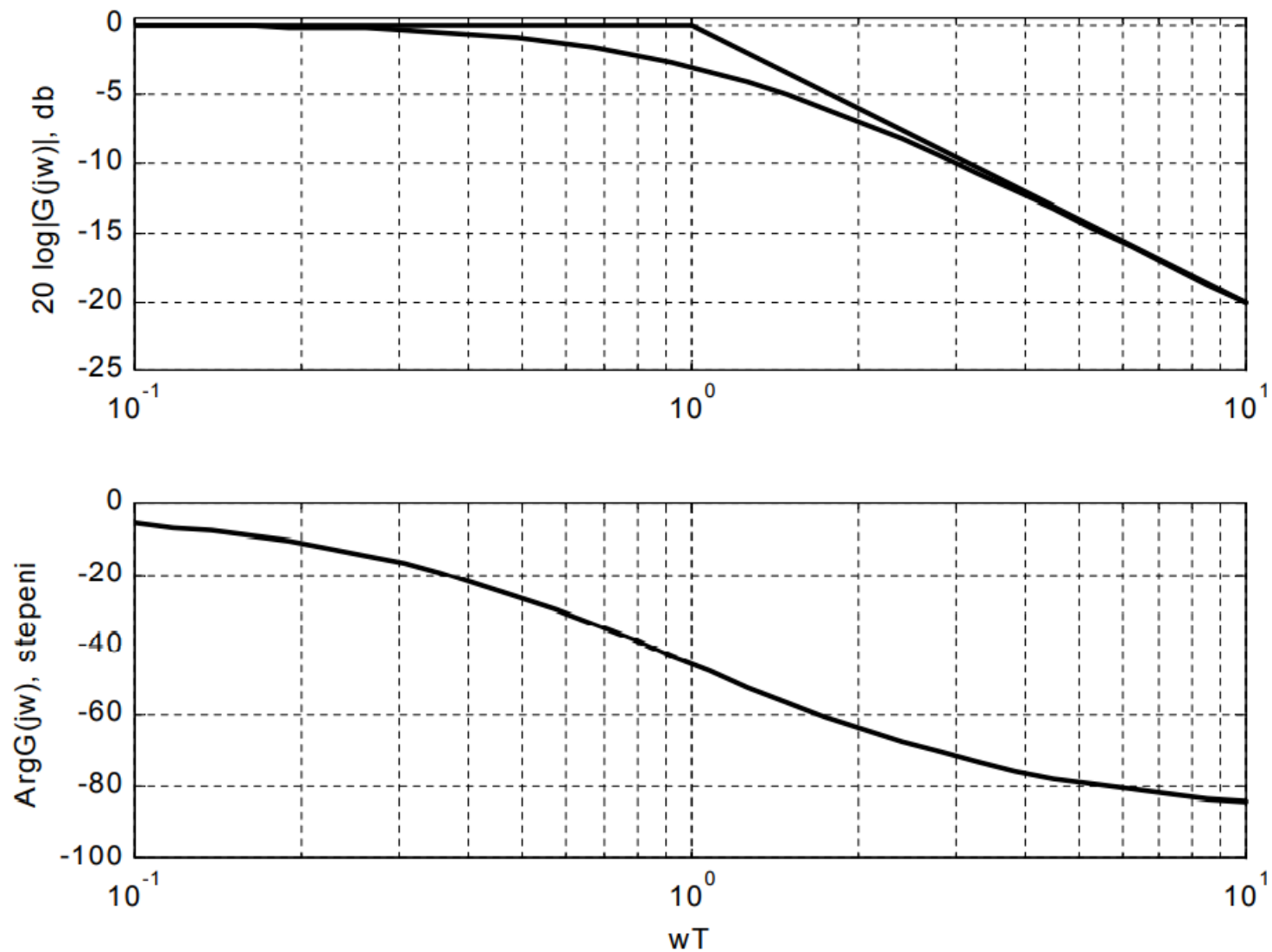
$$\text{Arg}\{G(j\omega)\} = 0 - \arctg \frac{\omega}{1} = -\arctg(\omega)$$

$$|G(j2)| = \frac{1}{\sqrt{4+1}} = \frac{1}{\sqrt{5}} \quad |G(j4)| = \frac{1}{\sqrt{16+1}} = \frac{1}{\sqrt{17}}$$

$$\text{Arg}\{G(j2)\} = -\arctg 2 = -1.107$$

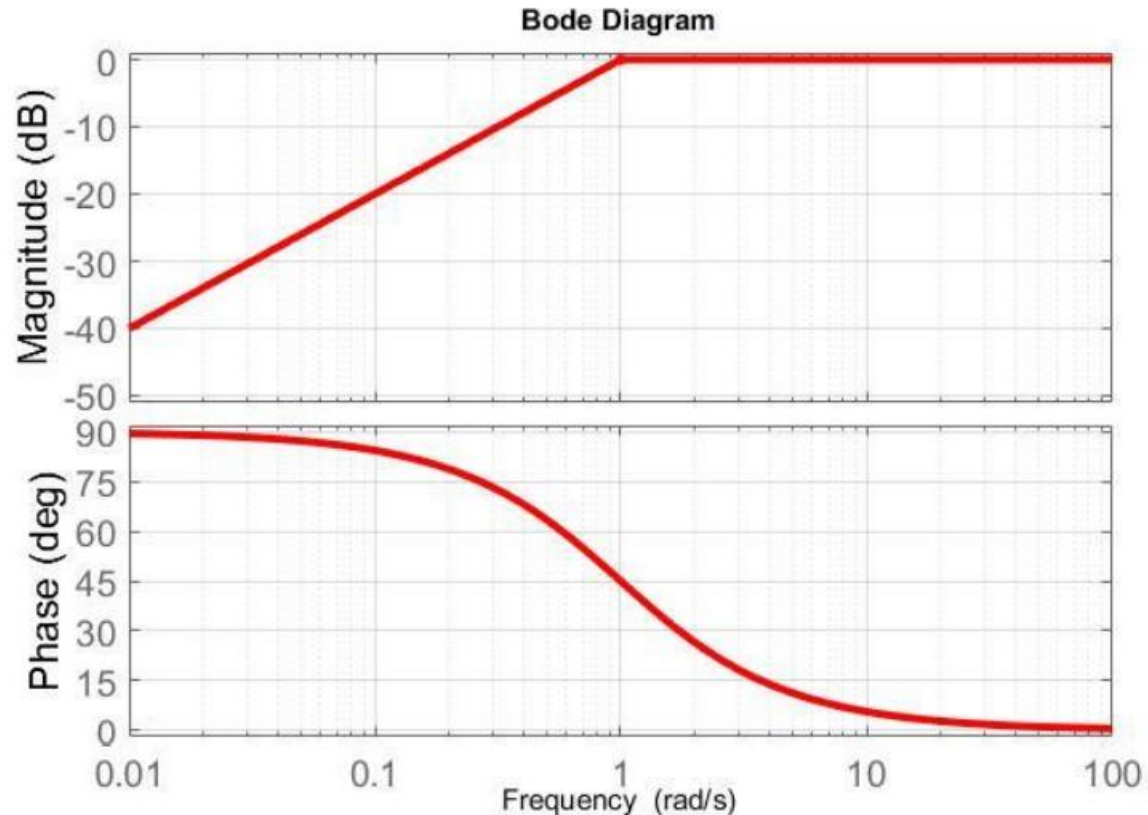
$$\text{Arg}\{G(j4)\} = -\arctg 4 = -1.3258$$

BODEOVI DIJAGRAMI



ZADATAK 4.

Na slici su prikazani Bodeovi dijagrami linearnog, stacionarnog sistema opisanog funkcijom prenosa $G(s)$. Odrediti odziv sistema u ustaljenom stanju ukoliko je pobudni signal $u(t) = \sin(2t) + 5 \cos(0.2t + \pi)$.



ZADATAK 4.

$$y(t) = y_1(t) + y_2(t)$$

$$y_1(t) = |G(2j)| \sin(2t + \text{Arg}\{G(2j)\})$$

$$y_2(t) = 5|G(0.2j)| \cos(0.2t + \pi + \text{Arg}\{G(0.2j)\})$$

$$|G(2j)|_{dB} = 0 \text{ dB} \longrightarrow |G(2j)| = 10^{\frac{1}{20} \cdot 0 \text{ dB}} = 1$$

$$|G(0.2j)|_{dB} \approx -14 \text{ dB} \longrightarrow |G(0.2j)| = 10^{\frac{1}{20} \cdot (-14) \text{ dB}} = 0.1995$$

$$\text{Arg}\{G(2j)\} \approx 26^\circ \longrightarrow \text{Arg}\{G(2j)\} = 26^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = 0.4538 \text{ rad}$$

$$\text{Arg}\{G(0.2j)\} \approx 78^\circ \longrightarrow \text{Arg}\{G(0.2j)\} = 78^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = 1.3613 \text{ rad}$$

ZADATAK 4.

$$y_1(t) = \sin(2t + 0.4538)$$

$$y_2(t) = 5 \cdot 0.1995 \cos(0.2t + \pi + 1.3613) = 0.9975 \cos(0.2t + 4.5029)$$

$$y(t) = \sin(2t + 0.4538) + 0.9975 \cos(0.2t + 4.5029)$$